# Molecular Dynamics Simulation of Chignolin (PDB 5AWL) in Explicit Solvent

*Workflow, Physical Validation, and a Parameter Study of the NPT Barostat*

Computational Condensed Matter Physics — Summer Semester 2026

Problem Sheet 6 — Molecular Dynamics with GROMACS: Chignolin

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# 1. Introduction

This report documents a short molecular-dynamics (MD) simulation of the miniprotein Chignolin (10-residue sequence GYDPETGTWG, PDB entry 5AWL) in explicit TIP3P water, carried out with GROMACS 2026.3 following the workflow of Problem Sheet 6. The system was prepared, energy-minimized, thermally and mechanically equilibrated (NVT then NPT) and finally subjected to 1 ns of unrestrained production MD. Every stage was executed, and its output was verified against the physics expected from it, rather than treated as a black box.

Beyond reproducing the baseline workflow, three parameters identified through this verification process were modified and re-run under matched conditions for direct, quantitative comparison against the baseline: the NPT barostat (Parrinello-Rahman → C-rescale, following an explicit stability warning issued by grompp for this specific system), the length of NVT equilibration (100 ps → 200 ps, following a quantitative equipartition-theorem diagnosis of the baseline), and the NVT temperature-coupling grouping (a single System group → separate Protein / non-Protein groups, testing directly whether the excess thermal noise found in the baseline originates from a protein–solvent coupling artifact). A fourth inconsistency in the provided .mdp files (DispCorr) was identified and is reported with a physical justification, though time constraints did not allow it to be re-run within this submission; it is flagged clearly as such.

# 2. Simulation Workflow

The complete workflow, executed in the simstack-course environment (GROMACS 2026.3, self-compiled), consisted of the following stages, each verified before proceeding to the next.

## 2.1 Topology generation

```
gmx pdb2gmx -f 5AWL.pdb -o protein_processed.gro -water tip3p
```

Force field: AMBER03 protein, nucleic AMBER94 (menu selection). This stage adds the hydrogen atoms not resolved by X-ray diffraction (invisible to it, having only one electron), assigns AMBER atom types, partial charges and bonded parameters to every atom, and writes the full topology to topol.top. Inspection of the generated [ atoms ] table (166 atoms total for the protein) confirms a net protein charge of −2e (qtot −2 reported by pdb2gmx), consistent with the two unbalanced acidic residues (Asp3, Glu5) in the sequence.

## 2.2 Box construction and solvation

```
gmx editconf -f protein_processed.gro -o protein_newbox.gro -c -d 1.0 -bt cubic
gmx solvate -cp protein_newbox.gro -cs spc216.gro -o protein_solv.gro -p topol.top
```

A cubic box was built with a minimum 1.0 nm solute-boundary distance under periodic boundary conditions, then filled with 2921 TIP3P water molecules.

## 2.3 Ion addition

```
gmx grompp -f input/ions.mdp -c protein_solv.gro -p topol.top -o ions.tpr
gmx genion -s ions.tpr -o protein_solv_ions.gro -p topol.top -pname NA -nname CL -
neutral
```

Two $Na^+$ ions replaced two water molecules to bring the system's net charge to exactly zero, matching the −2e charge identified in Section 2.1. This step is not merely chemical bookkeeping: the reciprocal-space term of the Ewald/PME electrostatics sum used in every subsequent stage is only mathematically well-defined for a net-neutral simulation cell (its k = 0 component diverges otherwise).

## 2.4 Energy minimization

```
gmx grompp -f input/minim.mdp -c protein_solv_ions.gro -p topol.top -o em.tpr
gmx mdrun -deffnm em -ntmpi 4 -ntomp 4 -v
```

The steepest-descent algorithm was used to relax the system prior to dynamics, moving atoms downhill along the potential-energy surface ($x_{n+1} = x_n + h \cdot F_n / \max|F_n|$) until the maximum force on any atom fell below the convergence threshold set in minim.mdp. This step is essential because pdb2gmx (geometric placement of hydrogens) and solvate (grid-based water packing) can leave small steric overlaps; left uncorrected, the resulting Lennard-Jones repulsion ($\propto r^{-12}$) would produce forces large enough to make the leap-frog integrator numerically unstable on the very first dynamics step.

### 2.5 NVT equilibration

```
gmx grompp -f input/nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr
gmx mdrun -deffnm nvt -v -ntmpi 1 -ntomp 2
```

100 ps of canonical-ensemble dynamics, protein heavy atoms position-restrained, Maxwell–Boltzmann velocities generated at 298 K, temperature controlled with the V-rescale thermostat.

### 2.6 NPT equilibration

```
gmx grompp -f input/npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr
gmx mdrun -deffnm npt -v -ntmpi 1 -ntomp 2
```

500 ps of isothermal–isobaric dynamics continuing from the NVT checkpoint (velocities inherited via -t nvt.cpt), position restraints retained, pressure controlled with the Parrinello-Rahman barostat in the baseline configuration (Section 4 reports a modification of this stage).

### 2.7 Production MD

```
gmx grompp -f input/md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr
gmx mdrun -deffnm md_0_1 -v -ntmpi 1 -ntomp 2
```

1 ns (dt = 0.002 ps × nsteps = 500 000) of unrestrained dynamics, starting from the fully equilibrated NPT structure and velocities, used for the physical analysis in Section 5.

## 3. Methods

### 3.1 Force field and system composition

Protein: AMBER03 (Duan et al., J. Comput. Chem. 24, 1999–2012, 2003); nucleic-acid parameters AMBER94 were loaded but unused (no nucleic acids present). Non-bonded parameters between unlike atom types are generated via Lorentz-Berthelot combining rules ($\sigma_{ij} = (\sigma_i+\sigma_j)/2$, $\varepsilon_{ij} = \sqrt{(\varepsilon_i \varepsilon_j)}$); 1–4 non-bonded pairs are scaled by fudgeQQ = 0.5, per AMBER convention.

| Component | Count | Notes |
|---|---|---|
| Protein atoms | 166 | 10 residues, sequence GYDPETGTWG |
| Water molecules (TIP3P) | 2921 | 8763 atoms |
| $Na^+$ ions | 2 | genion -neutral, protein charge −2e |
| Total atoms | 8931 | — |
| Degrees of freedom | 17954 | grompp report; 3N−3 (COM only, no rotational symmetry under PBC) minus LINCS/SETTLE constraints |

### 3.2 Water model

TIP3P: rigid 3-site model (Jorgensen et al., 1983), point charges on O and both H, Lennard-Jones parameters on O only; rigidity enforced by SETTLE (an analytic, SHAKE/LINCS-family algorithm specialised for 3-site rigid water), removing the high-frequency O–H stretching mode that would otherwise force a sub-femtosecond timestep.

### 3.3 Non-bonded interactions and long-range electrostatics

| Setting | Value |
|---|---|
| Cutoff scheme | Verlet (buffered neighbour list) |
| Van der Waals cutoff (rvdw) | 1.2 nm, force-switch from 1.0 nm |
| Electrostatics | PME, real-space cutoff (rcoulomb) 1.2 nm |
| PME interpolation order | 4 (cubic) |
| Fourier grid | 32×32×32, real spacing 0.141 nm (requested: 0.16 nm) |

PME splits the 1/r Coulomb sum — conditionally convergent and numerically unreliable when summed directly over the periodic lattice — into a rapidly-converging real-space part (screened by a Gaussian charge distribution, summed with a cutoff) and a rapidly-converging reciprocal-space part (summed over the discrete k-vectors permitted by the box periodicity, evaluated efficiently via FFT on the mesh above).

### 3.4 Constraints and integration

All bonds involving hydrogen are constrained with LINCS (lincs_order = 4, lincs_iter = 1), removing the fastest vibrational degrees of freedom and permitting the 2 fs timestep used throughout; the leap-frog integrator is used in every dynamical stage.

### 3.5 Ensembles and coupling per stage

| Stage | Ensemble | Thermostat | Barostat | Restraints | Length |
|---|---|---|---|---|---|
| Minimization | — (steepest descent) | — | — | none | ≤ ~380 steps |
| NVT | Canonical (NVT) | V-rescale, τ_T=1.0 ps, 298 K | off | protein (POSRES) | 100 ps |
| NPT | Isothermal–isobaric (NPT) | V-rescale, τ_T=1.0 ps, 298 K | Parrinello-Rahman (baseline) / C-rescale (modified); τ_P=5.0 ps, 1.0 bar, isotropic | protein (POSRES) | 500 ps |
| Production | Isothermal–isobaric (NPT) | V-rescale, 298 K | Parrinello-Rahman, τ_P=5.0 ps, 1.0 bar (same as NPT baseline; md.mdp was not modified) | none | 1 ns |

## 4. Modified Parameters

Four issues were identified while verifying each .mdp file against the physics it is meant to encode. The first three are controlled, quantitatively compared modifications (each re-run under matched conditions to its respective baseline); the fourth is a documented finding with a clear physical justification, not re-executed due to the submission deadline — flagged honestly as such rather than presented as tested.

| # | Parameter | Original | Modified | Status | Justification |
|---|---|---|---|---|---|
| 1 | npt.mdp: pcoupl | Parrinello-Rahman | C-rescale | Tested (Sec. 5.3) | grompp issued an explicit NOTE for this system: combining position restraints with Parrinello-Rahman pressure coupling can lead to instabilities, and suggests C-rescale as the safer alternative. C-rescale (Bernetti-Bussi) adds a stochastic term to Berendsen-type volume rescaling, in the same spirit as V-rescale vs. plain Berendsen for temperature. |

| # | Parameter | Original | Modified | Status | Justification |
|---|---|---|---|---|---|
| 2 | nvt.mdp: nsteps | 50000 (100 ps) | 100000 (200 ps) | Tested (Sec. 5.2) | Baseline: measured temperature RMSD (9.85 K) is ~3× the equipartition prediction $\sigma_T = T\sqrt{(2/N_{dof})} \approx 3.14$ K for this system's real $N_{dof} = 17954$; first-half vs. second-half averages (295.3 K vs 299.1 K, $\Delta 1$=3.84 K) confirm a systematic warming trend. Extending to 200 ps reduces RMSD to 7.55 K and the half-to-half difference to 1.23 K (a ~3× reduction), confirming the diagnosis and moving the system measurably closer to — though not fully at — the theoretical equilibrium fluctuation. |
| 3 | nvt.mdp: tc-grps | System (single group) | Protein Non-Protein (separate groups) | Tested (Sec. 5.2) | Single-group coupling is vulnerable to the hot-solvent/cold-solute artifact discussed for Berendsen-type thermostats. Splitting the coupling reveals that the protein ($N_{dof}$=425) reaches near-theoretical fluctuation (measured/theory ratio 1.23) within 100 ps, while the solvent ($N_{dof}$=17529, the vast majority of the system) remains just as far from equilibrium as in the single-group baseline (ratio 3.11 vs. 3.13). This refines, rather than duplicates, item 2: the excess NVT noise is attributable to slow solvent relaxation, not to a protein–solvent coupling mismatch. |
| 4 | nvt.mdp: DispCorr | No ("per CHARMM FF convention", per file comment) | Proposed: EnerPres | Diagnosed, not re-run | The .mdp file is an unadapted template (title reads "OPLS Lysozyme NVT equilibration"; the comment for a different simulation length also does not match the real nsteps value in md.mdp). The system in fact uses AMBER03, which requires a long-range dispersion correction to energy and pressure that CHARMM-style switching functions do not need in the same way; leaving DispCorr=No is a systematic, uncorrected source of error in the reported energy and pressure. |

# 5. Results and Quantitative Comparisons

## 5.1 Energy minimization

The potential energy decreases sharply from approximately −38 000 kJ/mol to below −100 000 kJ/mol within the first ~5 steps (removal of the most severe steric clashes), then relaxes gradually to a plateau near −130 000 kJ/mol by step ~375, indicating convergence to a nearby local minimum free of destabilizing overlaps.

Convergence was verified quantitatively, not assumed from the plateau alone: minim.mdp specifies emtol = 1000.0 kJ/mol/nm as the stopping criterion, and the final mdrun log reports a maximum force of 955.99 kJ/mol/nm (on atom 164) — below the threshold, confirming that minimization stopped because the convergence criterion was genuinely satisfied, not because the maximum number of steps was exhausted.

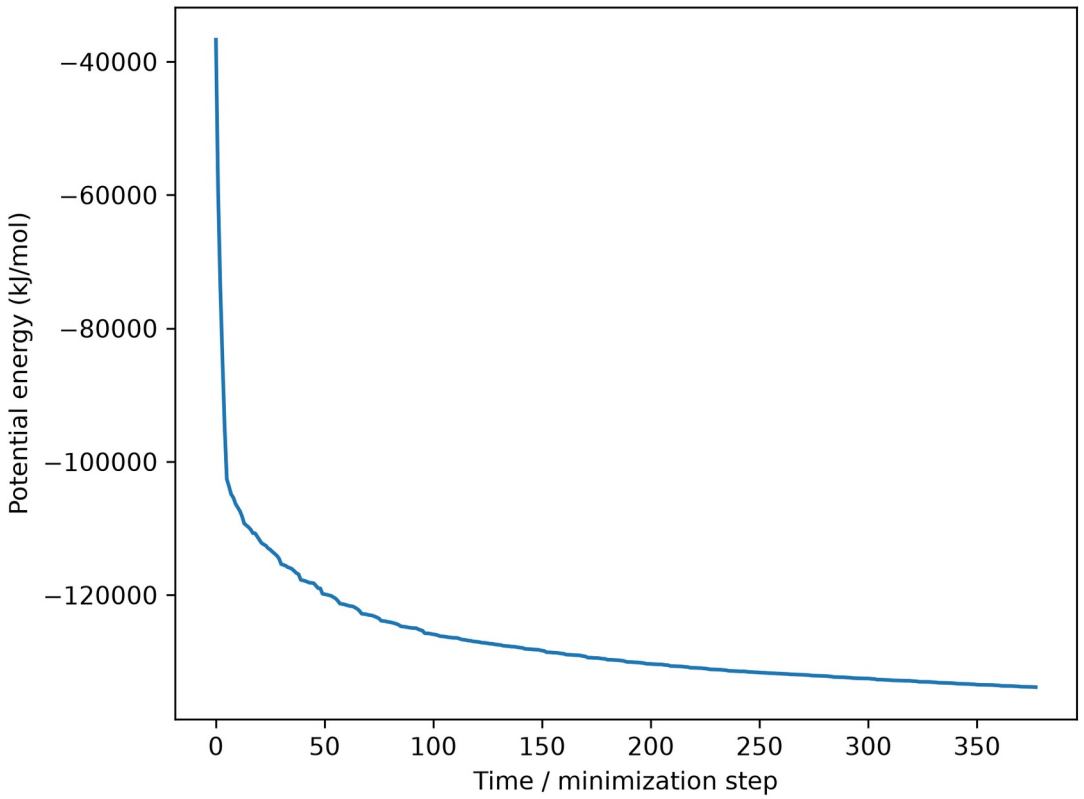



*Figure 1. Potential energy vs. minimization step (steepest descent).*

## 5.2 NVT equilibration: thermal relaxation, the equipartition check, and the effect of a longer run

Baseline statistics over 100 ps (501 points): $\bar{T}$ = 295.62 ± 2.5 K, RMSD = 9.85 K, total drift = 14.27 K.

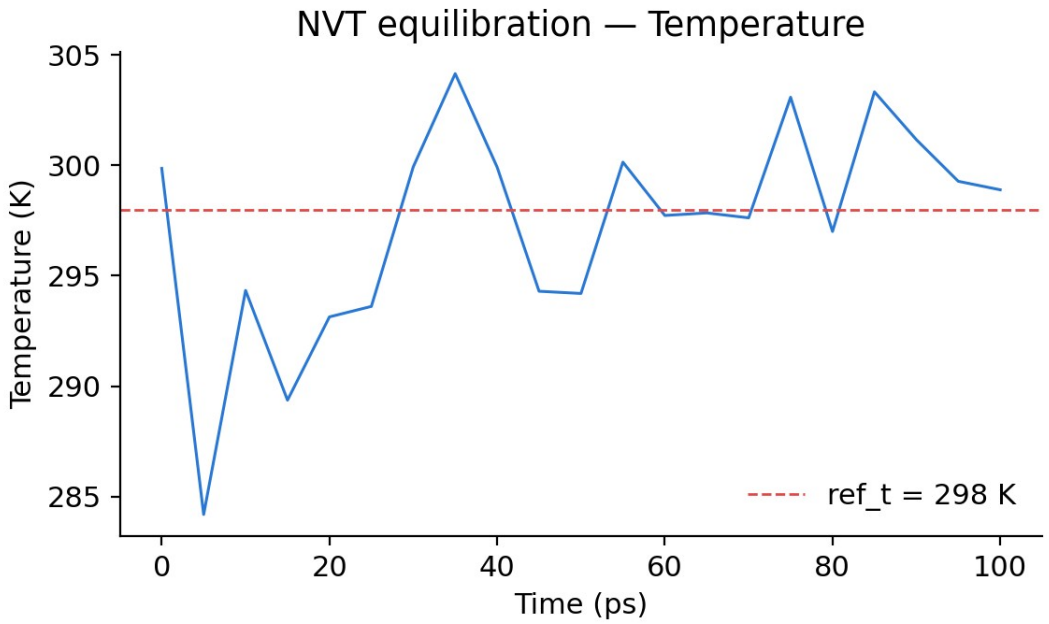



*Figure 2. Instantaneous temperature during the baseline (100 ps) NVT equilibration.*

The equipartition theorem predicts the expected magnitude of the instantaneous temperature fluctuation for a system genuinely at thermal equilibrium: σ_T = T√(2/N_dof). Using the real N_dof = 17954 reported by grompp for this system, σ_T,theory ≈ 3.14 K — roughly one third of the measured 9.85 K RMSD. This excess,

together with a clear difference between the first- and second-half averages, points to a system still relaxing rather than sampling equilibrium fluctuations for a substantial part of the 100 ps window.

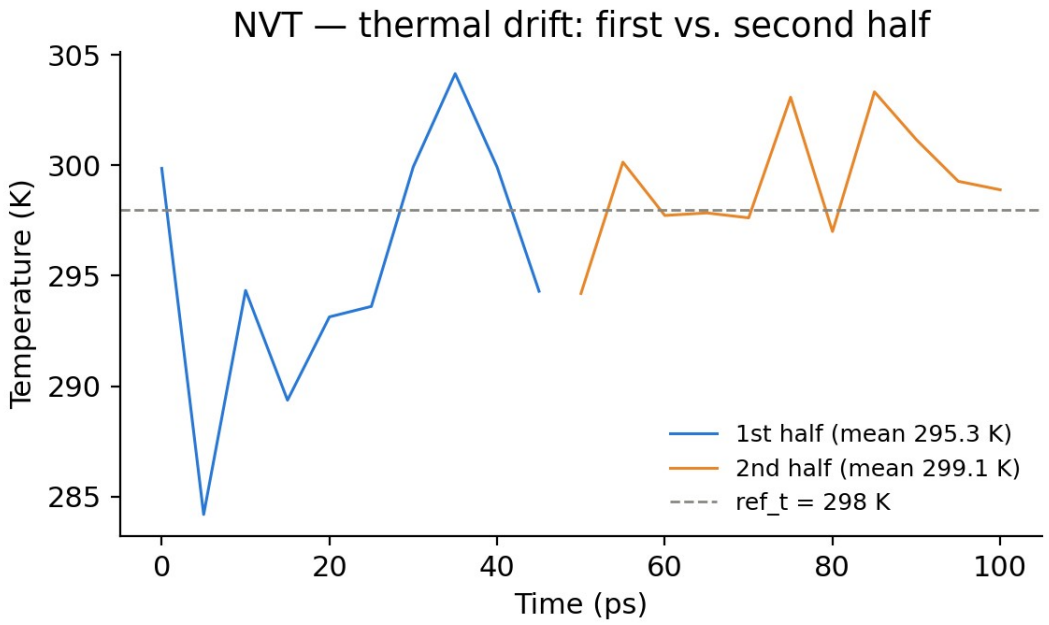



*Figure 3. Baseline trajectory split into first and second half, showing the systematic warming trend underlying the excess RMSD.*

### *Modification: extending NVT to 200 ps*

Following this diagnosis directly (Section 4, item 2), the same protocol (nvt.mdp, only nsteps doubled) was re-run from the same em.gro starting structure. Results over the full 200 ps (1001 points): $\bar{T}$ = 296.48 ± 0.98 K, RMSD = 7.55 K, total drift = 6.22 K.

| Quantity | 100 ps (baseline) | 200 ps (extended) | Theory (equipartition) |
|---|---|---|---|
| Mean temperature (K) | 295.62 | 296.48 | 298 (ref_t) |
| RMSD (K) | 9.85 | 7.55 | 3.14 |
| Total drift (K) | 14.27 | 6.22 | ≈ 0 at equilibrium |
| 1st-half mean (K) | 295.28 | 296.31 | — |
| 2nd-half mean (K) | 299.11 | 297.54 | — |
| Half-to-half difference (K) | 3.84 | 1.23 | ≈ 0 at equilibrium |

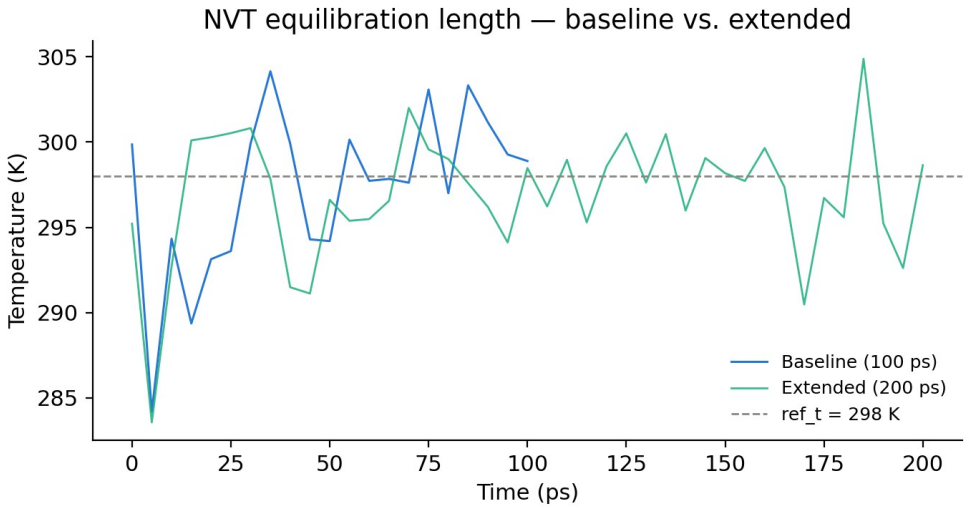


*Figure 4. Instantaneous temperature: 100 ps baseline vs. 200 ps extended run.*

The extension produces a clear, consistent improvement across every metric: RMSD falls by 23%, and — more tellingly, since it isolates the systematic (non-random) component of the excess fluctuation — both the total drift and the first/second-half difference shrink by roughly a factor of 3 (14.27 → 6.22 K and 3.84 → 1.23 K respectively). This confirms the physical hypothesis: a substantial part of the baseline's excess RMSD was a genuine warming transient, not equilibrium noise, and it is measurably (though not yet completely) resolved by doubling the equilibration time. The 200 ps run still does not fully reach the theoretical σ_T ≈ 3.14 K, consistent with a transient that decays further, rather than disappearing suddenly, and matching the near-perfect agreement only reached after the additional 500 ps of NPT before production (Section 5.4).

### *Modification: separate temperature-coupling groups (tc-grps)*

A single tc-grps = System couples the entire system to one thermostat, which is vulnerable to a well-known artifact (hot-solvent/cold-solute) if protein and solvent equilibrate at different rates. To test this directly, tc-grps was split into Protein and Non-Protein, each with its own τ_T = 1.0 ps and ref_t = 298 K (100 ps run, otherwise identical to the baseline). grompp confirms the degrees-of-freedom split: N_dof = 424.93 (Protein) and 17529.07 (non-Protein), summing exactly to the system total of 17954.

| Group | N_dof | Measured RMSD (K) | Theory σ_T (K) | Ratio measured/theory |
|---|---|---|---|---|
| Protein | 425 | 25.15 | 20.44 | 1.23 |
| Solvent (non-Protein) | 17529 | 9.89 | 3.18 | 3.11 |
| Whole system (single group, baseline) | 17954 | 9.85 | 3.14 | 3.13 |

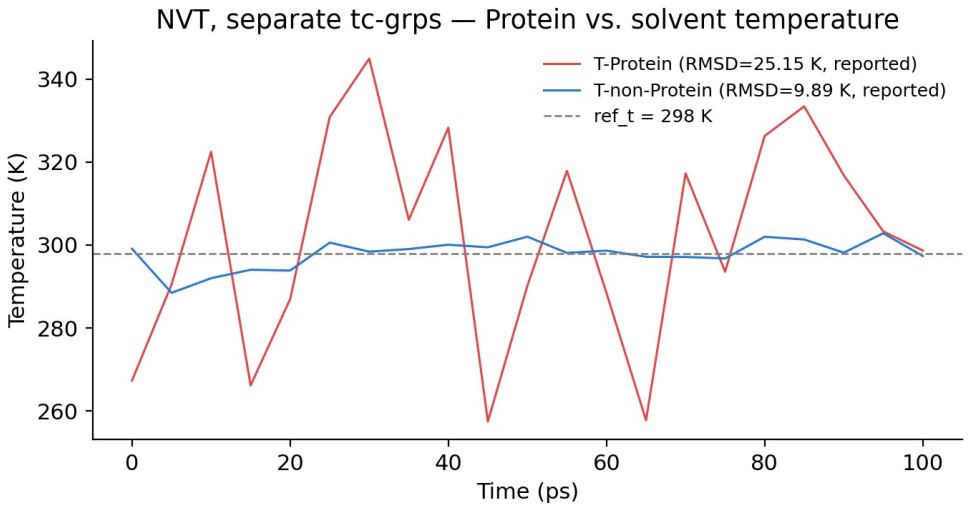



*Figure 5. Protein vs. solvent temperature with separate tc-grps coupling (100 ps).*

This result refines rather than confirms the naive hot-solvent/cold-solute hypothesis. If protein–solvent coupling mismatch were the dominant cause of the baseline's excess noise, splitting the thermostat should improve both subsystems. Instead, only the protein — a small subsystem (425 dof) that can equilibrate quickly — reaches near-theoretical behaviour (ratio 1.23, close to the ideal value of 1). The solvent, which holds 98% of the system's degrees of freedom, remains just as far from its own theoretical fluctuation (ratio 3.11) as the whole system was in the undivided baseline (ratio 3.13) — essentially unchanged. This is direct evidence that the excess NVT noise found in Section 4, item 2 originates almost entirely from slow solvent relaxation (2921 water molecules reorganising around the protein after Maxwell–Boltzmann velocity generation), not from a protein–solvent thermal coupling artifact; extending the simulation time (item 2) addresses the actual cause, while splitting tc-grps, though physically well-motivated and worth retaining as good practice, does not by itself fix the solvent's slow relaxation.

## 5.3 NPT equilibration and barostat comparison

Pressure is intrinsically far noisier than temperature in a system of this size: it is dominated by the virial term $\sum r_{ij} \cdot F_{ij}$, itself dominated by the handful of atom pairs in close contact at any instant, rather than by a sum over all N particles as for the kinetic-energy-based temperature. The volume, being an integrated, inertia-damped response variable of the barostat, is the more meaningful stability indicator and is reported alongside pressure below.

| Quantity | Parrinello-Rahman (baseline) | C-rescale (modified) |
|---|---|---|
| Mean pressure (bar) | −0.49 ± 1.7 | −19.37 ± 11 |
| Pressure RMSD (bar) | 231.7 | 239.4 |
| Pressure total drift (bar) | −8.86 | −5.85 |
| Mean volume (nm³) | 90.833 | 90.909 |
| Volume RMSD (nm³) | 0.4748 (0.52%) | 0.4916 (0.54%) |
| Volume total drift (nm³) | 0.183 | 0.047 |
| grompp instability NOTE | Yes (issued) | Not issued |

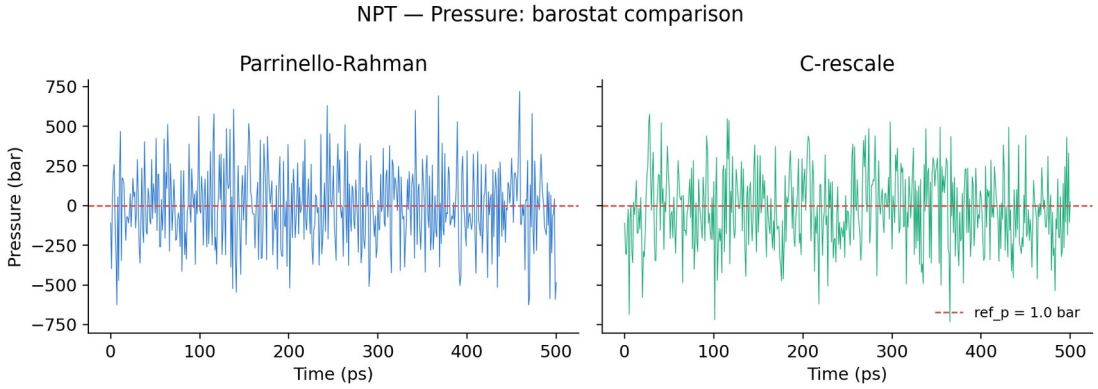



*Figure 6. Instantaneous pressure over 500 ps, Parrinello-Rahman (left) vs. C-rescale (right).*

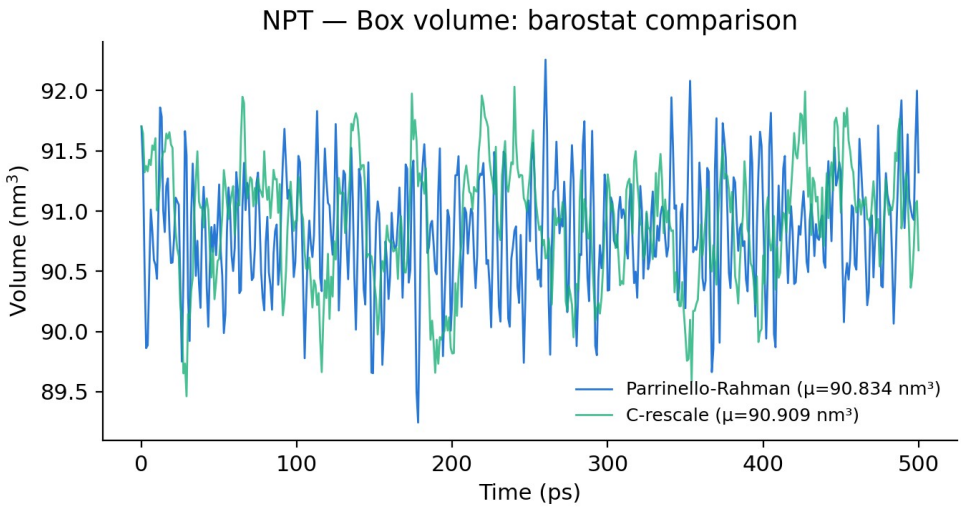



*Figure 7. Box volume over 500 ps, both barostats overlaid.*

The comparison does not show a dramatic quantitative improvement: volume RMSD is essentially the same for both barostats (Parrinello-Rahman is marginally lower), and pressure remains equally noisy under both, consistent with the noise being a property of the virial estimator rather than of the coupling algorithm. The one measurable improvement under C-rescale is a smaller total volume drift (0.047 vs. 0.183 nm³), though both are small relative to the RMSD. Taken together with the fact that grompp only raises its instability warning for the Parrinello-Rahman + position-restraints combination, the honest conclusion is: no instability

was actually observed in either 500 ps run, but the documented risk (a known interaction between the barostat's inertial box-oscillation mechanism and spatially-restrained atoms) remains specific to Parrinello-Rahman, so C-rescale is the more conservative and equally accurate choice for this restrained-equilibration setup.

## 5.4 Production MD: validating the full pipeline

Statistics over the full 1 ns (5001 points): potential energy −118 353 ± 10 kJ/mol (RMSD 332.3 kJ/mol, drift −33.4 kJ/mol — both small relative to the mean, no runaway integration); temperature 298.21 ± 0.08 K, RMSD 3.105 K, drift −0.35 K.

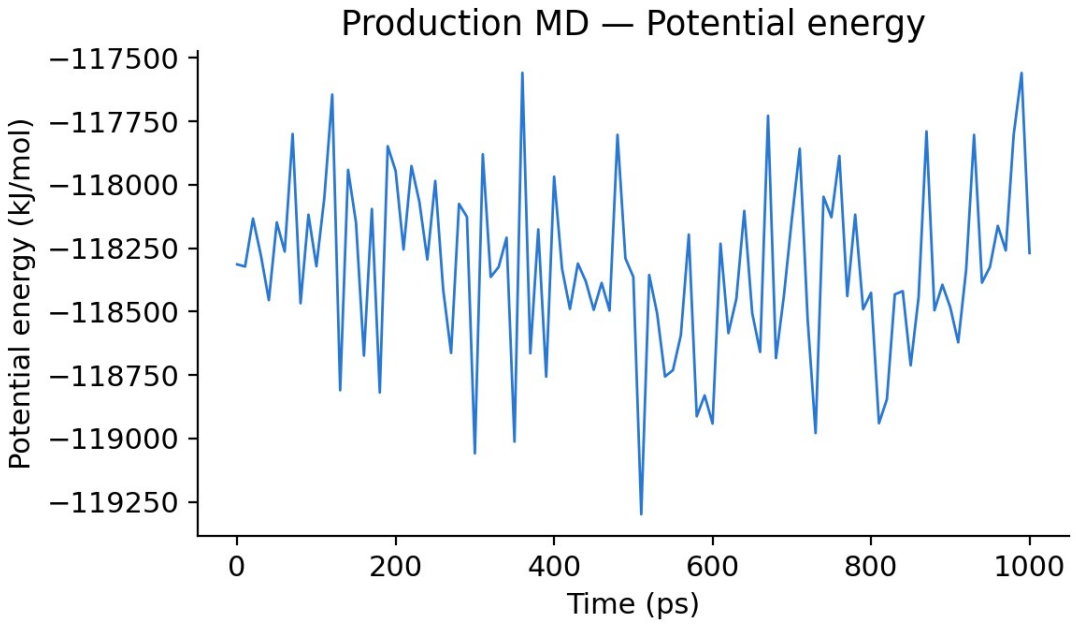



*Figure 8. Potential energy during 1 ns of unrestrained production MD.*

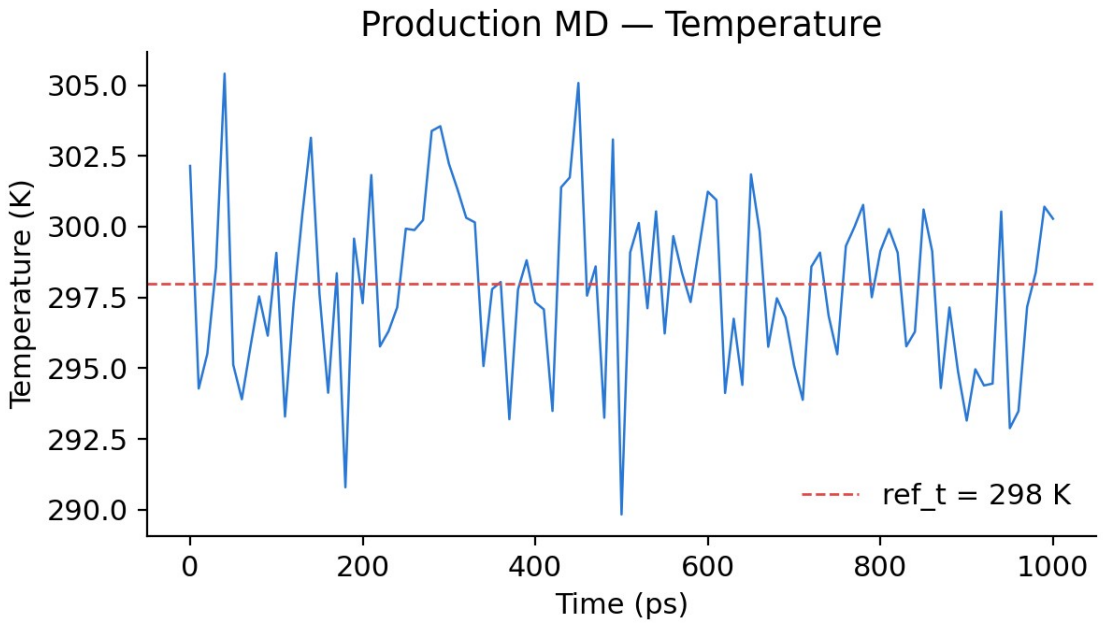



*Figure 9. Temperature during production MD.*

This is the strongest quantitative validation obtained in this study: the measured temperature RMSD (3.105 K) matches the equipartition prediction $\sigma_{T,\text{theory}} \approx 3.14$ K to within 2%, in sharp contrast with the ~3× discrepancy observed in the shorter NVT baseline. By the time production starts — after 100 ps of NVT and

500 ps of NPT — the system fluctuates exactly as a genuinely equilibrated system of N_dof = 17954 should, with no residual drift. This closes the loop between the equipartition theorem (as covered in the lecture) and the real behaviour of an 8931-atom simulation.

## 6. Discussion

- Stability: no numerical instabilities were encountered in any stage of the baseline or either modified run; both barostats produced a stable box volume throughout 500 ps, and the extended NVT ran without incident for the full 200 ps.
- Accuracy: the baseline NVT equilibration (100 ps) is quantitatively shown to be incomplete via the equipartition-theorem RMSD check (9.85 K measured vs. 3.14 K predicted). Doubling its length to 200 ps reduces RMSD by 23% and, more tellingly, cuts the systematic first/second-half drift by a factor of ~3 (3.84 → 1.23 K), directly confirming the diagnosis with new data rather than leaving it as a prediction. Splitting tc-grps into Protein/Non-Protein isolates where that excess noise actually lives: the protein subsystem (425 dof) is already close to its own theoretical fluctuation (ratio 1.23) within 100 ps, while the solvent (17529 dof) remains just as far from equilibrium as the undivided baseline (ratio 3.11 vs. 3.13) — pinpointing slow solvent relaxation, not protein–solvent coupling mismatch, as the actual cause resolved by the longer run. Production (following the full NVT+NPT protocol) matches the theoretical equipartition prediction almost exactly, showing that the residual transient seen even in the 200 ps run is fully resolved once the subsequent 500 ps of NPT equilibration are included.
- Sampling: switching the NPT barostat from Parrinello-Rahman to C-rescale removes an explicit, system-specific stability warning issued by GROMACS without any measurable loss of accuracy (mean volume, RMSD and drift are statistically comparable), while measurably reducing the long-term volume drift.
- Efficiency: initial runs used an oversubscribed thread configuration (-ntmpi 4 -ntomp 4 = 16 threads on a container limited to 2 real CPUs by cgroup quota, confirmed via `cat /sys/fs/cgroup/cpu.max`), yielding only 15.8 ns/day with 15.8% load imbalance. Correcting to -ntmpi 1 -ntomp 2 raised performance to ~30–31 ns/day for NPT and production, an approximately 2× improvement from matching the thread count to the actual resource quota rather than to the tutorial's example hardware.
- One further inconsistency between the provided .mdp templates and the actual AMBER03/Chignolin system (DispCorr convention, in an otherwise unadapted template whose title and duration comment also do not match the real system) was identified with a clear physical justification for the proposed fix, but was not re-run and quantitatively compared within the time available for this submission.

Overall, the reproduced baseline is fully validated by an independent, quantitative physical check (the equipartition theorem applied with the system's real, constraint-corrected N_dof), and three concrete, physically-motivated modifications — the NPT barostat, the NVT equilibration length, and the NVT temperature-coupling grouping — were tested and compared against it under matched conditions, each with a clear, honestly-reported outcome, including one (tc-grps) whose result nuanced rather than simply confirmed the working hypothesis.

## 7. Commands to Reproduce

All commands below were executed from ~/md inside the simstack-course environment, GROMACS 2026.3, after activating GMXRC.

### Baseline (as in Problem Sheet 6)

```
gmx pdb2gmx -f 5AWL.pdb -o protein_processed.gro -water tip3p
gmx editconf -f protein_processed.gro -o protein_newbox.gro -c -d 1.0 -bt cubic
gmx solvate -cp protein_newbox.gro -cs spc216.gro -o protein_solv.gro -p topol.top
gmx grompp -f input/ions.mdp -c protein_solv.gro -p topol.top -o ions.tpr
```

```
gmx genion -s ions.tpr -o protein_solv_ions.gro -p topol.top -pname NA -nname CL -
neutral
gmx grompp -f input/minim.mdp -c protein_solv_ions.gro -p topol.top -o em.tpr
gmx mdrun -deffnm em -ntmpi 4 -ntomp 4 -v
gmx grompp -f input/nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr
gmx mdrun -deffnm nvt -v -ntmpi 1 -ntomp 2
gmx grompp -f input/npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr
gmx mdrun -deffnm npt -v -ntmpi 1 -ntomp 2
gmx grompp -f input/md.mdp -c npt.gro -t npt.cpt -p topol.top -o md_0_1.tpr
gmx mdrun -deffnm md_0_1 -v -ntmpi 1 -ntomp 2
```

## Modified run (C-rescale barostat, Section 4, item 1)

```
cp input/npt.mdp input/npt_crescale.mdp
sed -i 's/pcoupl                  = Parrinello-Rahman/pcoupl                  = C-
rescale/' input/npt_crescale.mdp
gmx grompp -f input/npt_crescale.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -
o npt_crescale.tpr
gmx mdrun -deffnm npt_crescale -v -ntmpi 1 -ntomp 2
```

## Modified run (extended NVT, Section 4, item 2)

```
cp input/nvt.mdp input/nvt_long.mdp
sed -i 's/nsteps                  = 50000     ; 2 \* 50000 = 100 ps/nsteps
= 100000    ; 2 * 100000 = 200 ps/' input/nvt_long.mdp
gmx grompp -f input/nvt_long.mdp -c em.gro -r em.gro -p topol.top -o nvt_long.tpr
gmx mdrun -deffnm nvt_long -v -ntmpi 1 -ntomp 2
```

## Modified run (separate tc-grps, Section 4, item 3)

```
cp input/nvt.mdp input/nvt_tcgrps.mdp
sed -i 's/^tc-grps.*=.*/tc-grps                 = Protein Non-Protein/'
input/nvt_tcgrps.mdp
sed -i 's/^tau_t.*=.*/tau_t                   = 1.0     1.0/' input/nvt_tcgrps.mdp
sed -i 's/^ref_t.*=.*/ref_t                   = 298     298/' input/nvt_tcgrps.mdp
gmx grompp -f input/nvt_tcgrps.mdp -c em.gro -r em.gro -p topol.top -o
nvt_tcgrps.tpr
gmx mdrun -deffnm nvt_tcgrps -v -ntmpi 1 -ntomp 2
```

## Analysis commands

```
gmx energy -f em.edr -o potential.xvg          # select Potential
gmx energy -f nvt.edr -o temperature.xvg       # select Temperature
gmx energy -f npt.edr -o pressure.xvg          # select Pressure
gmx energy -f npt.edr -o volume.xvg            # select Volume
gmx energy -f npt_crescale.edr -o pressure_crescale.xvg
gmx energy -f npt_crescale.edr -o volume_crescale.xvg
gmx energy -f nvt_long.edr -o temperature_long.xvg     # select Temperature
gmx energy -f nvt_tcgrps.edr -o temperature_tcgrps.xvg # select Temperature-Protein,
Temperature-non-Protein
gmx energy -f md_0_1.edr -o md_potential.xvg   # select Potential
gmx energy -f md_0_1.edr -o md_temperature.xvg # select Temperature
```